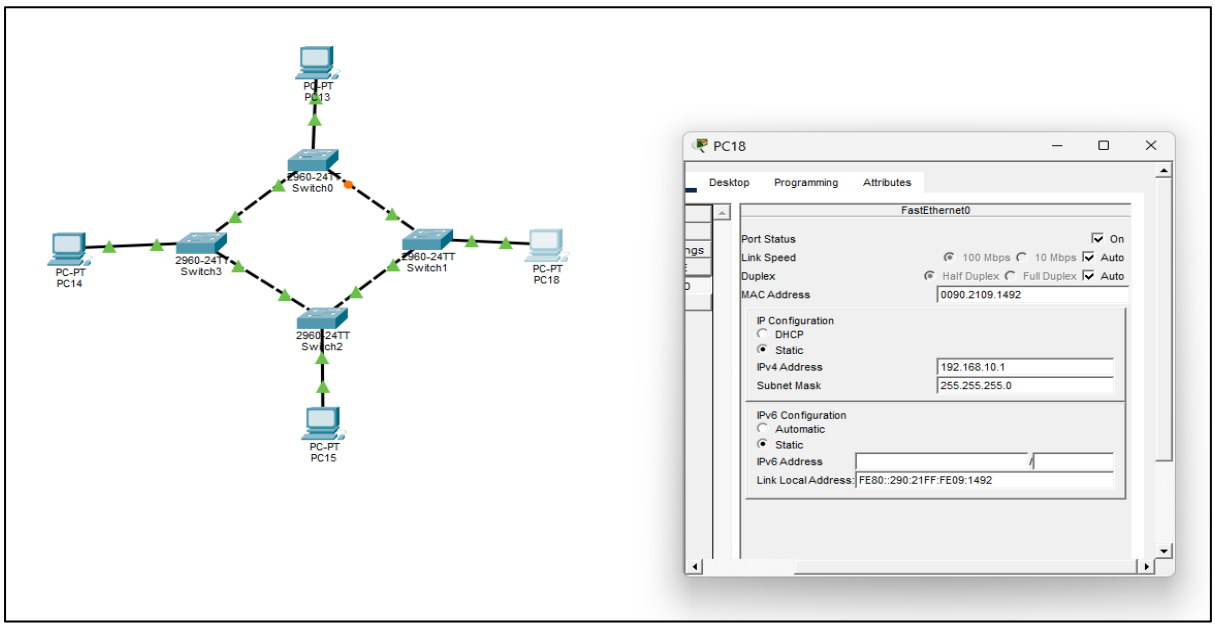
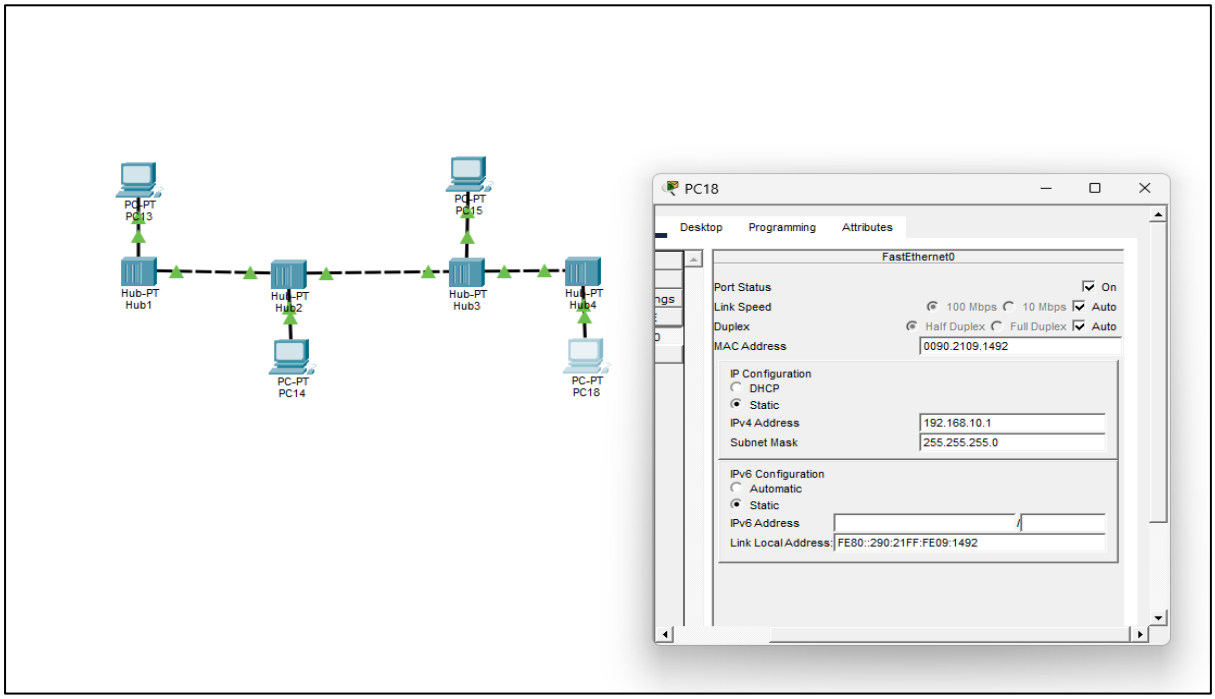
EXP-Network Topologies Using CISCO Packet Tracer

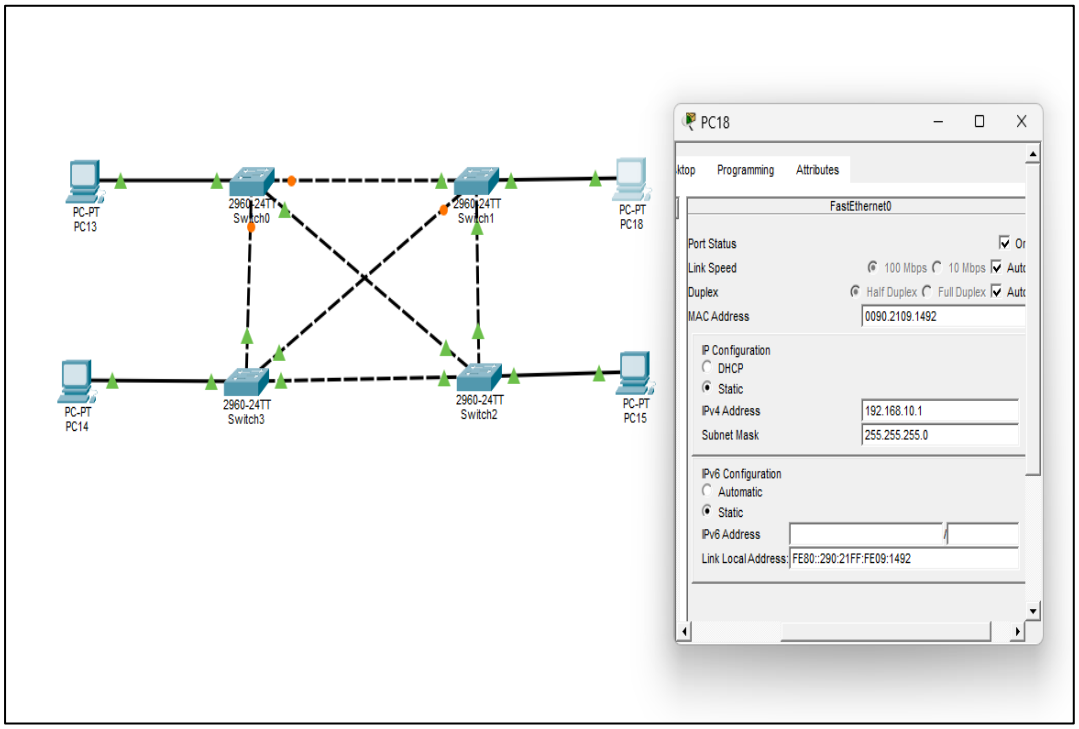
1. Ring Topology :



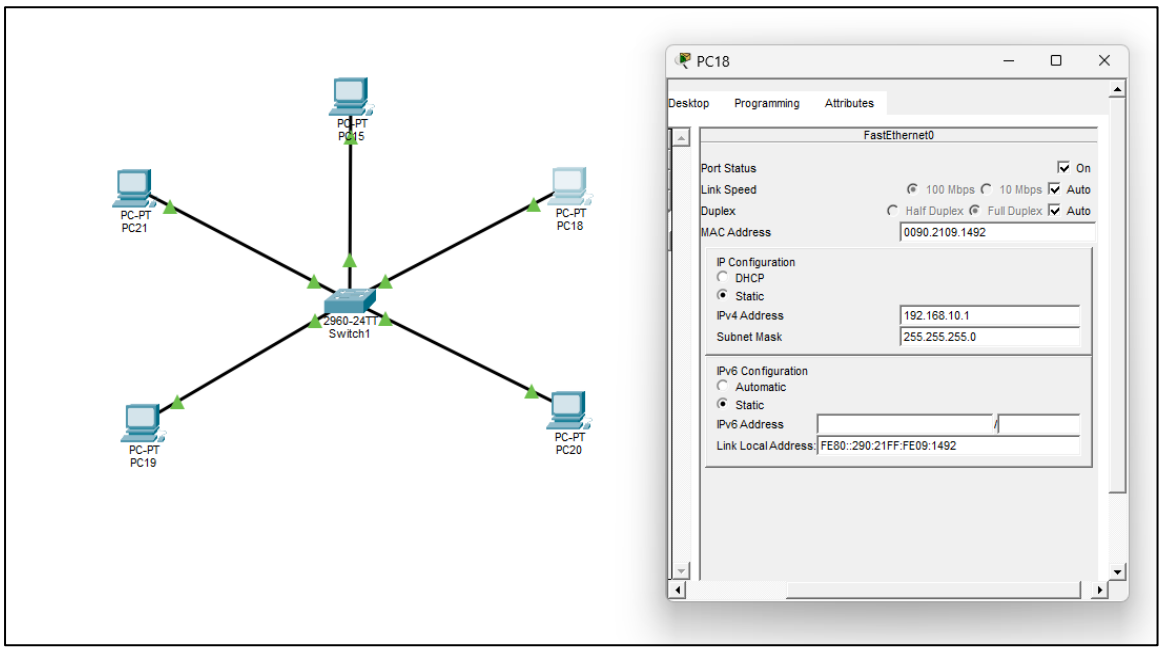
2. Bus Topology :



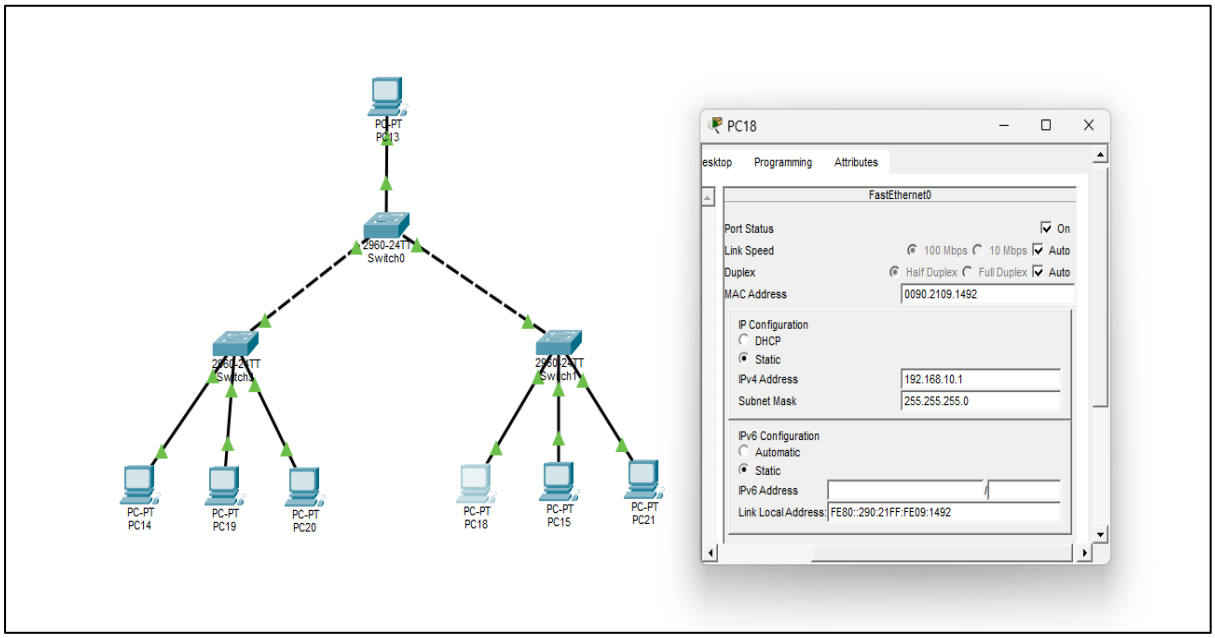
3. Mesh Topology :



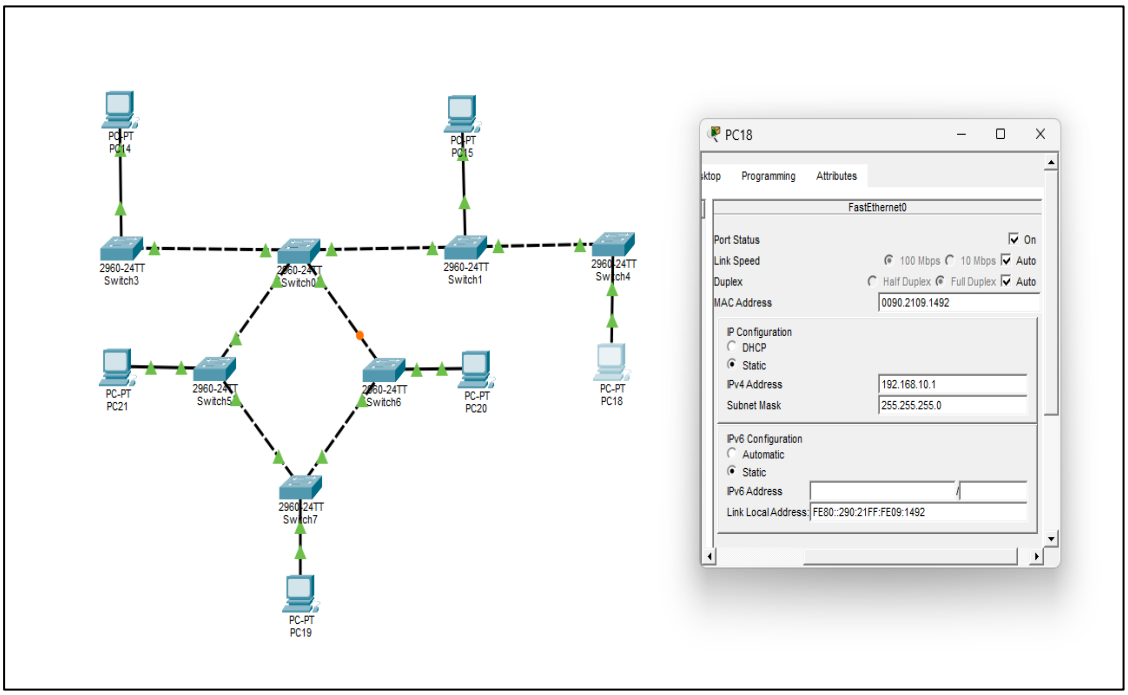
4. Star Topology :



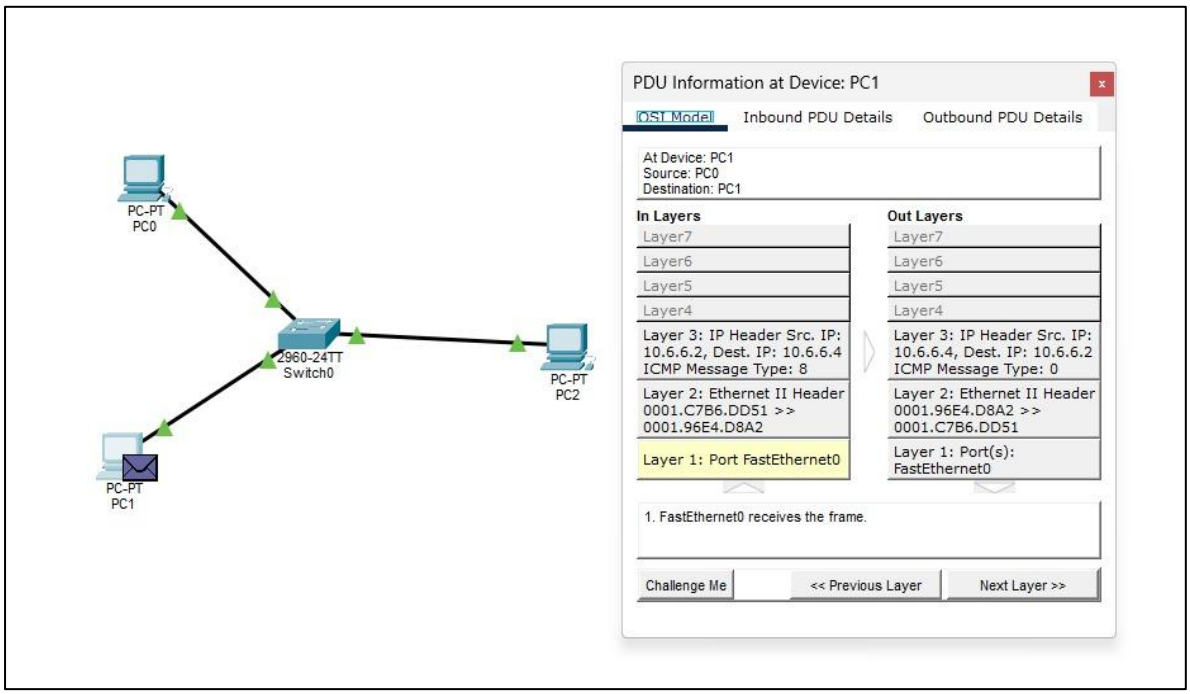
5. Tree Topology :

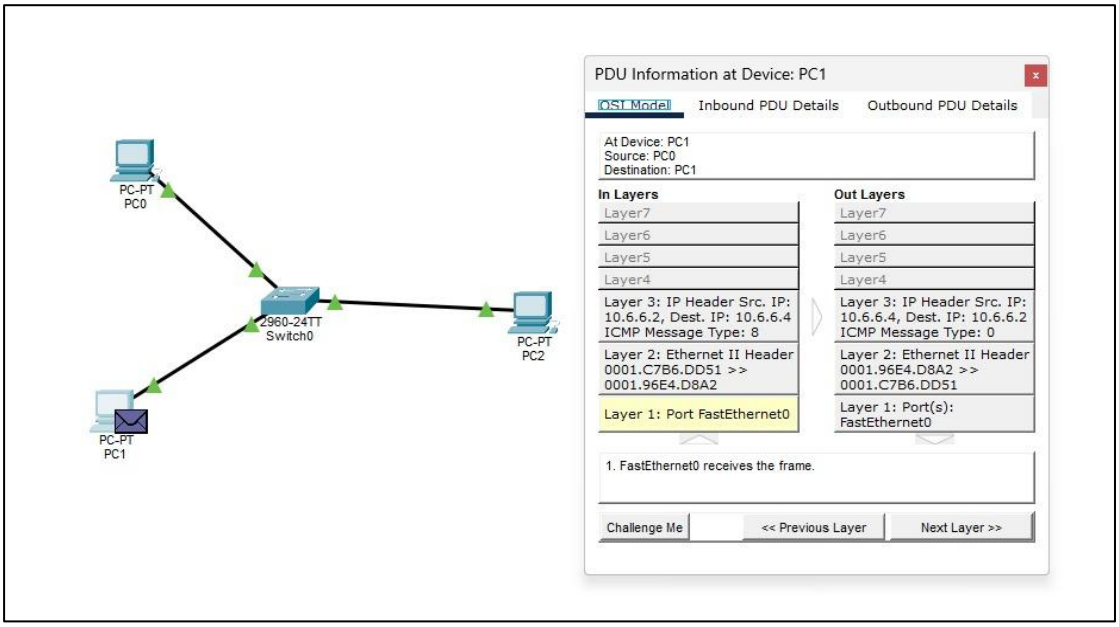


6. Hybrid Topology :

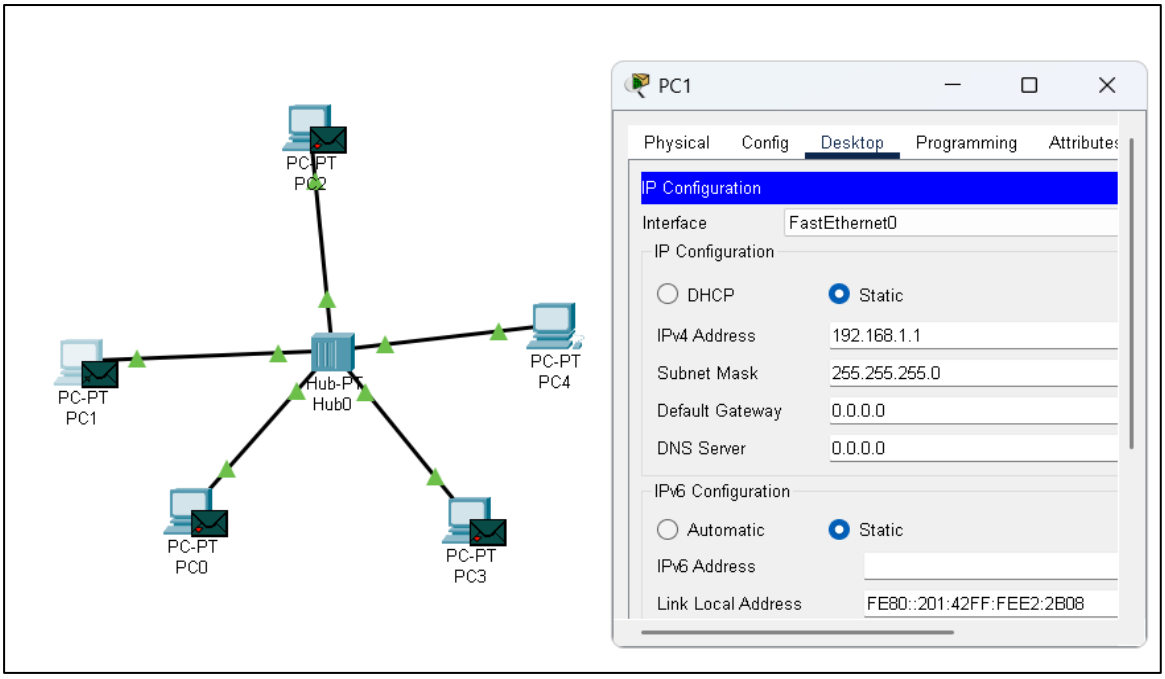


Data Link Layer Traffic Simulation (ARP analysis)

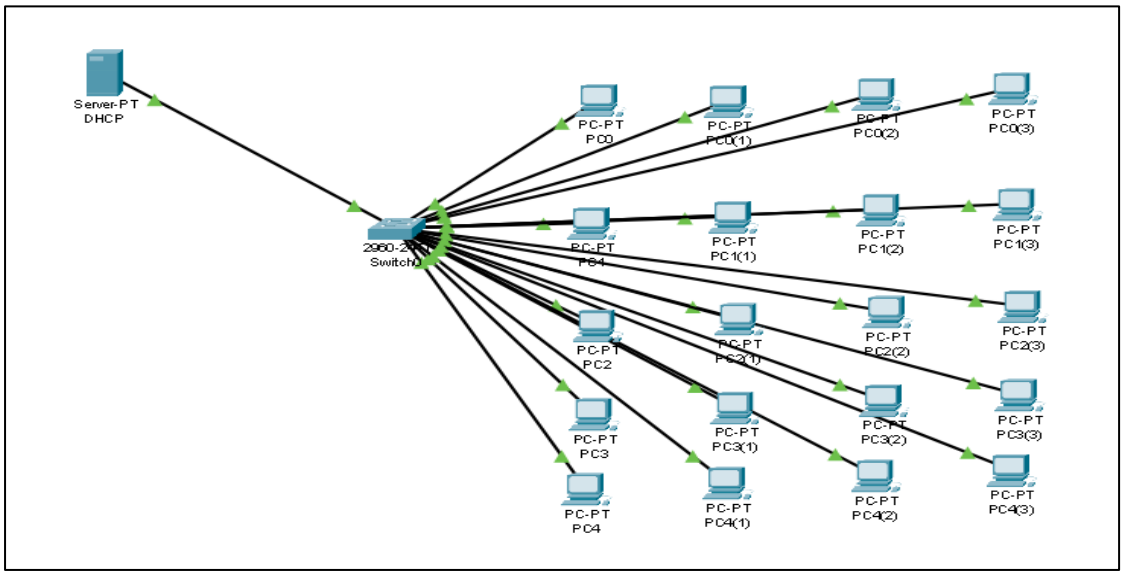




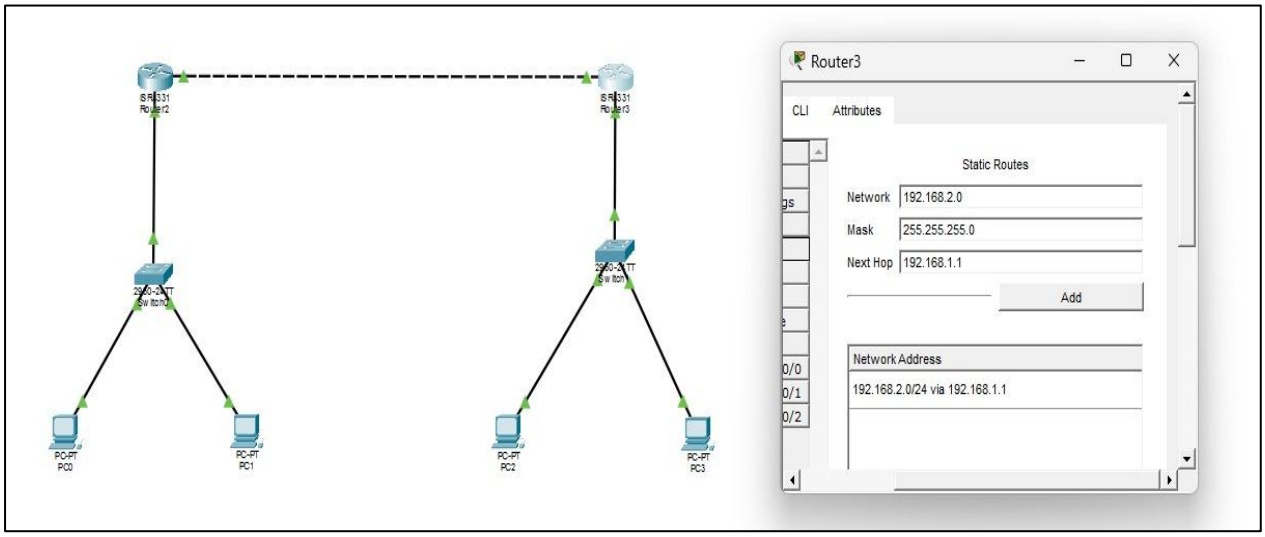
EXP- Data Link Traffic Simulation using Packet Tracer (CSMA/CD & CSMA/CA)

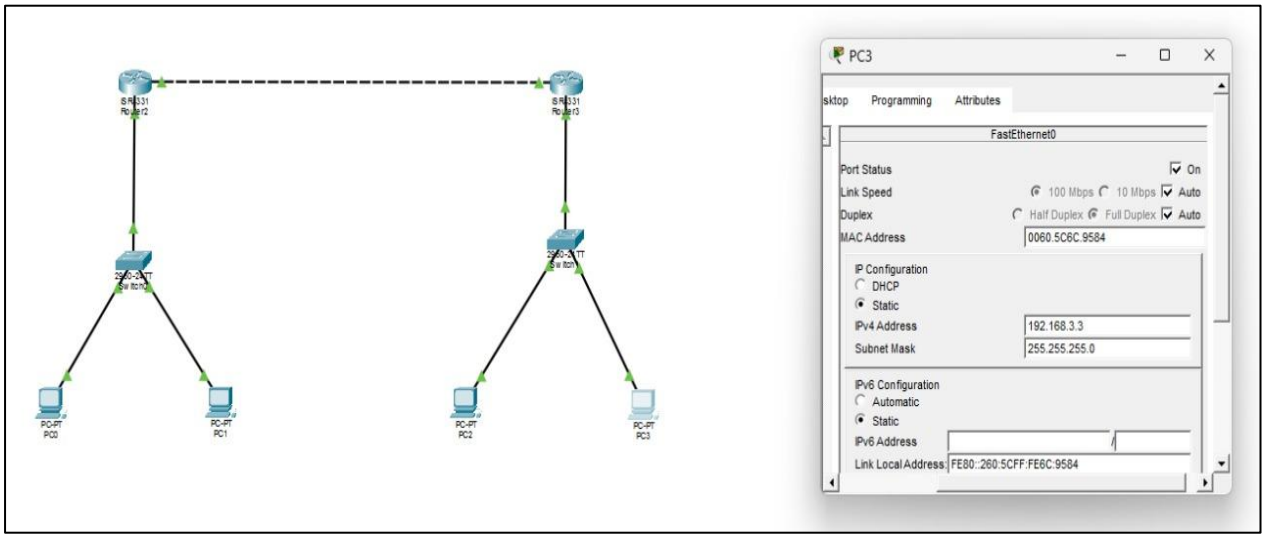


EXP-Creating a Computer Lab Network using Packet Tracer

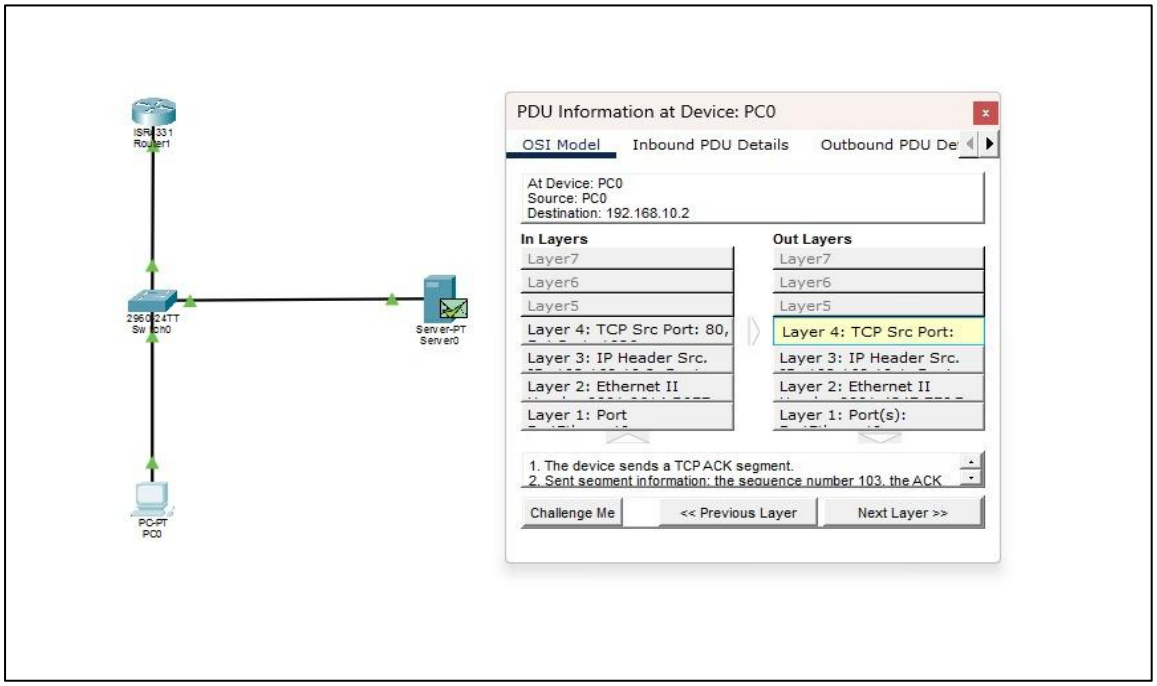


EXP- Configuration of a Simple Static Routing in Packet Tracer

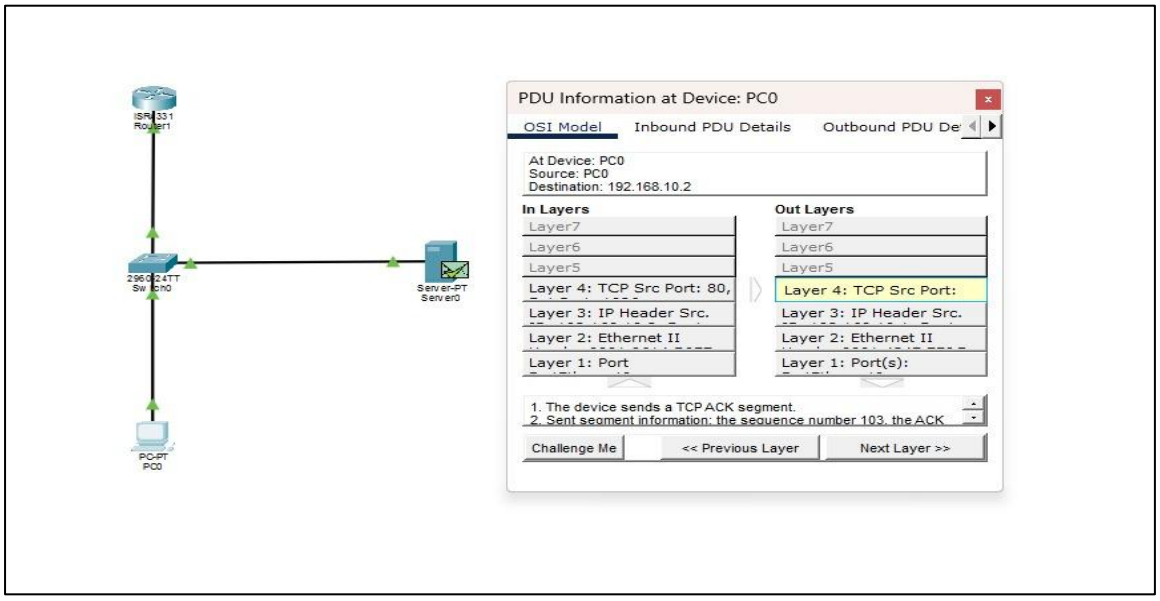




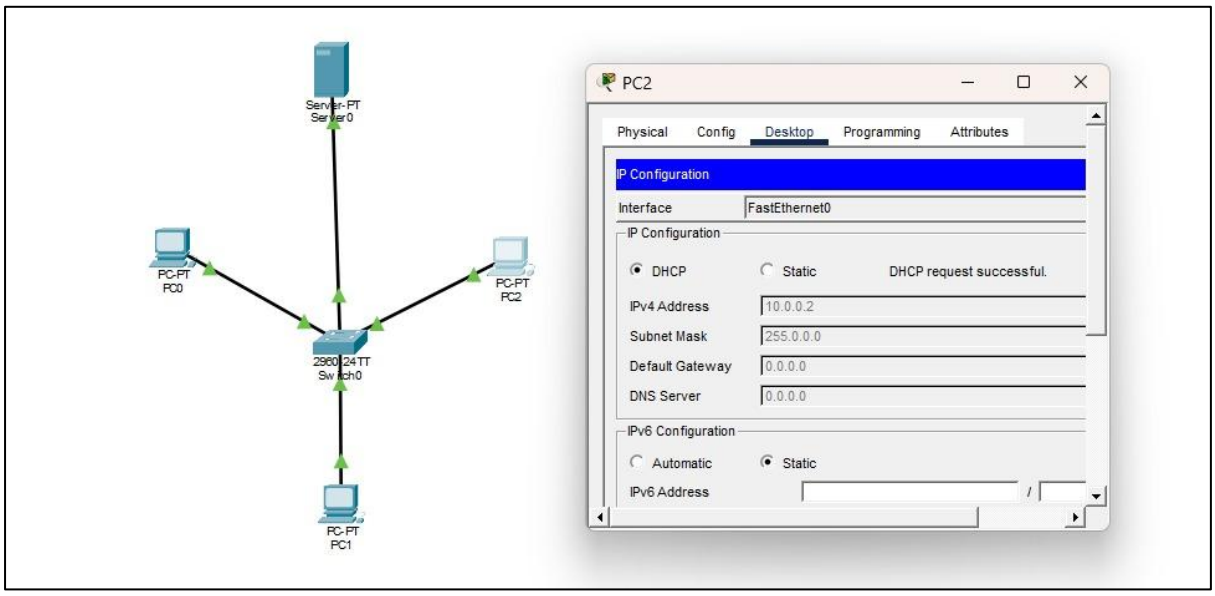
EXP-Design the Functionalities and Exploration of TCP using Packet Tracer



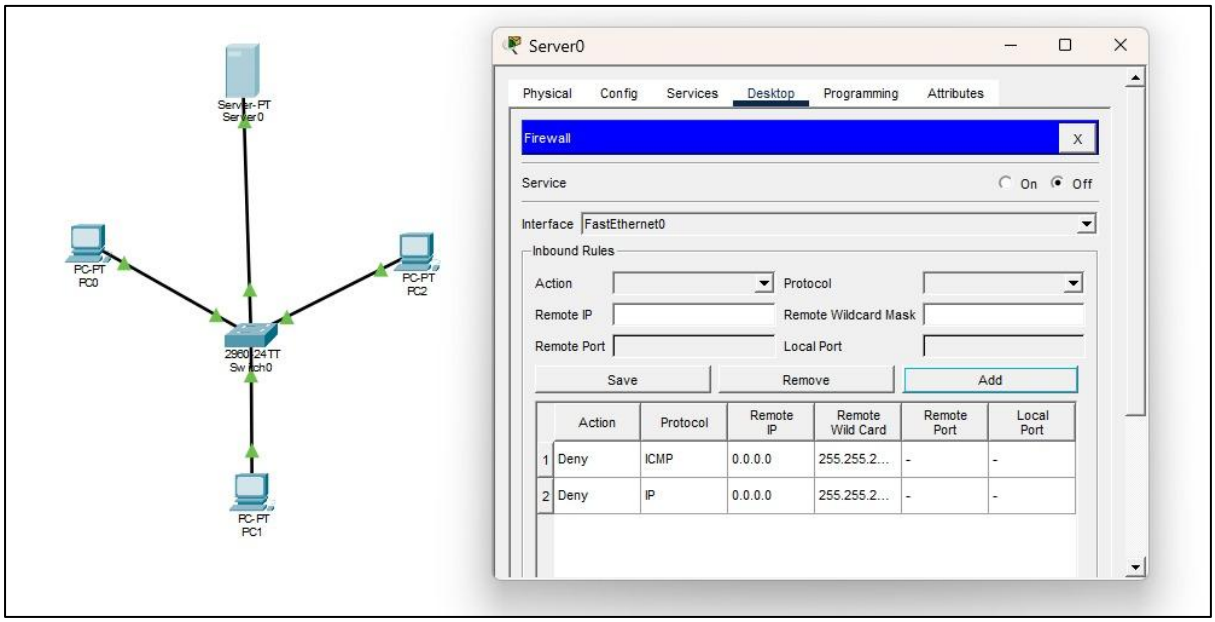
EXP-Networking Model for Subnetting- Class C Addressing

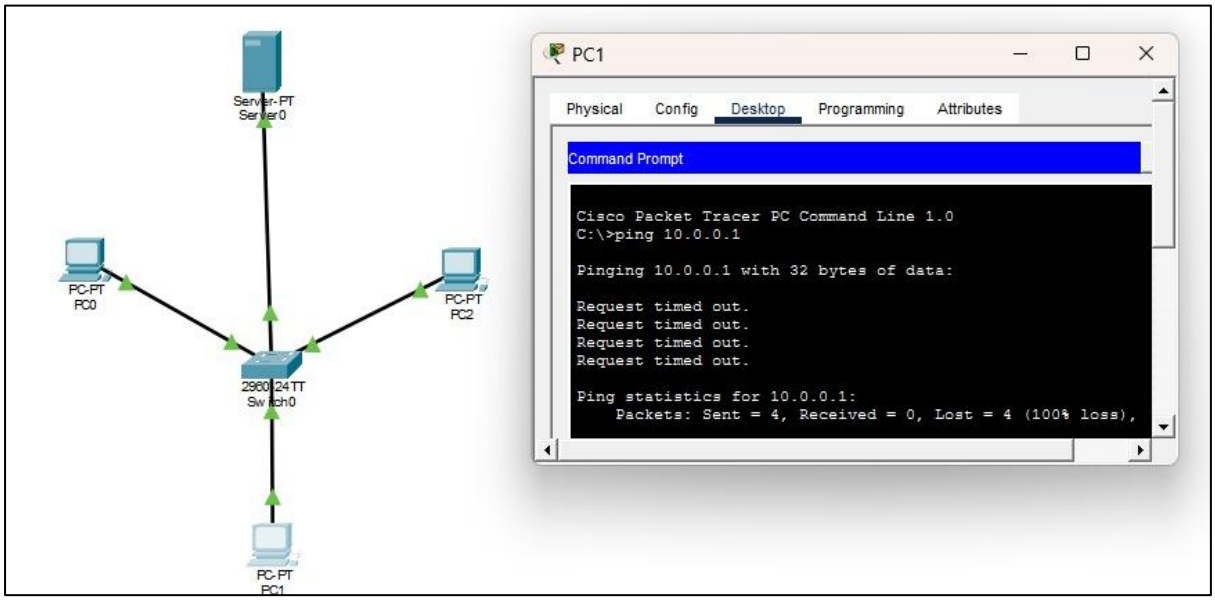


EXP- Configuration of DHCP in Packet Tracer

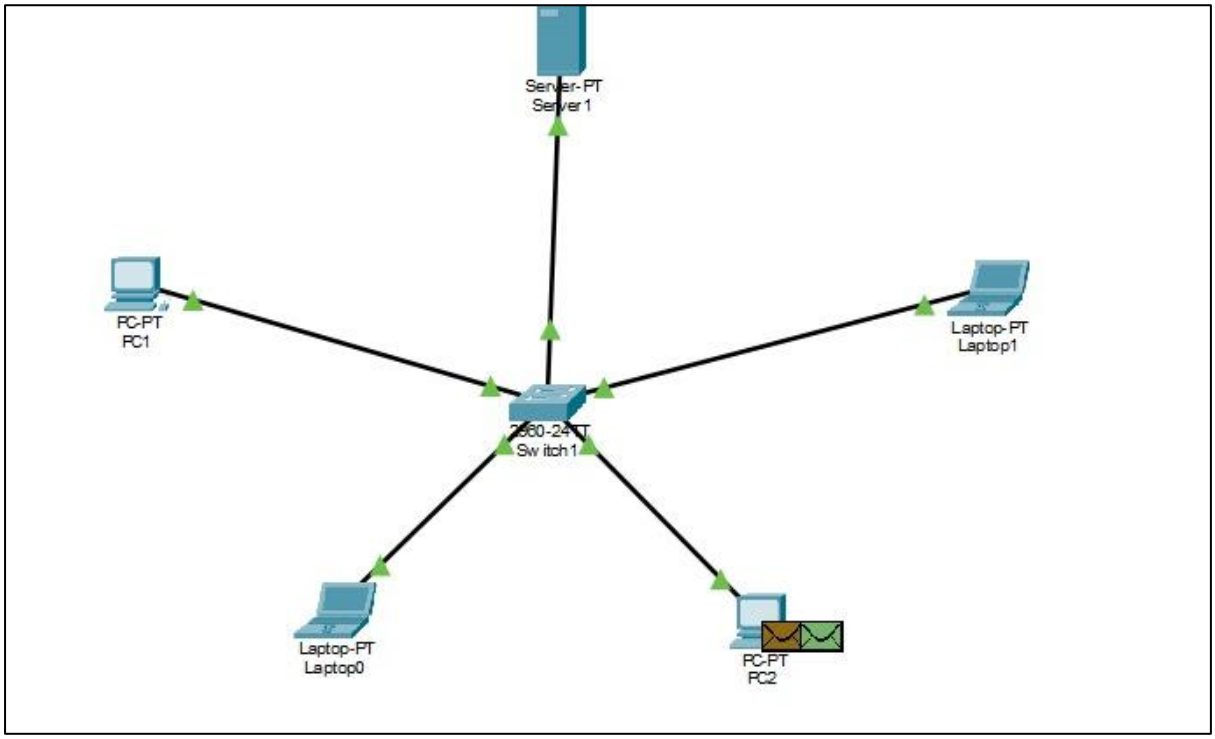


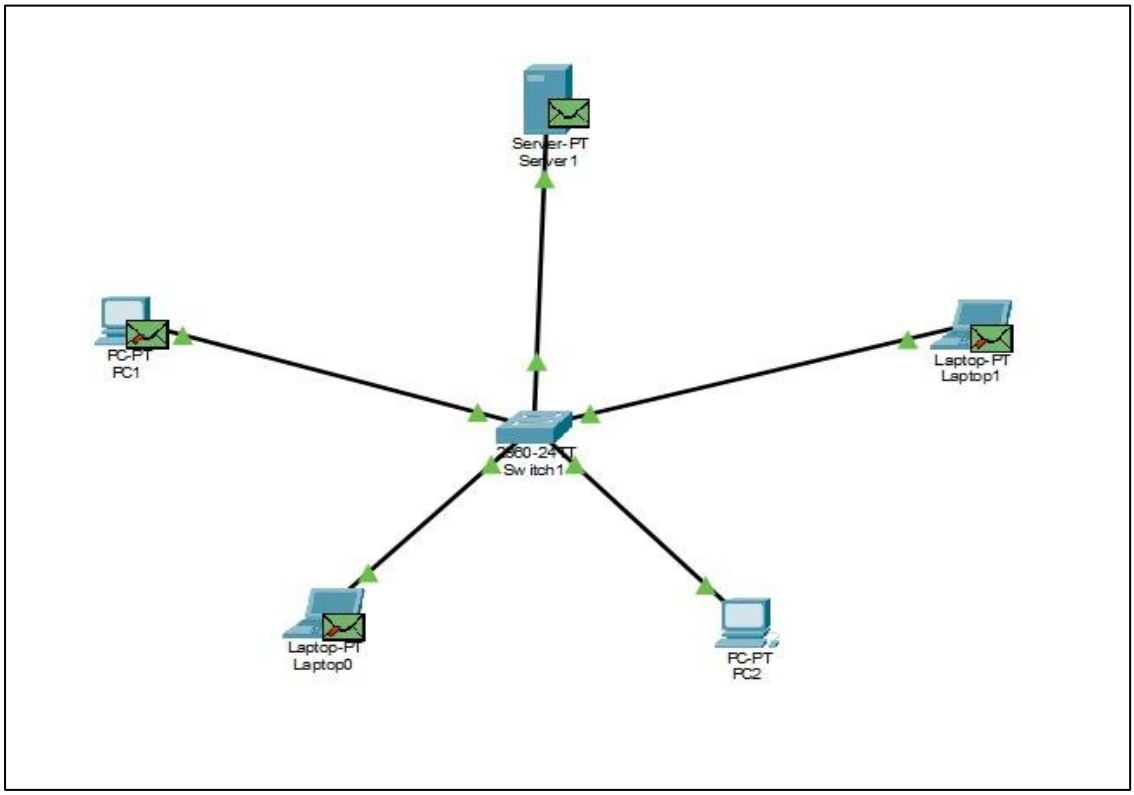
EXP- Configuration of Firewall in Packet Tracer



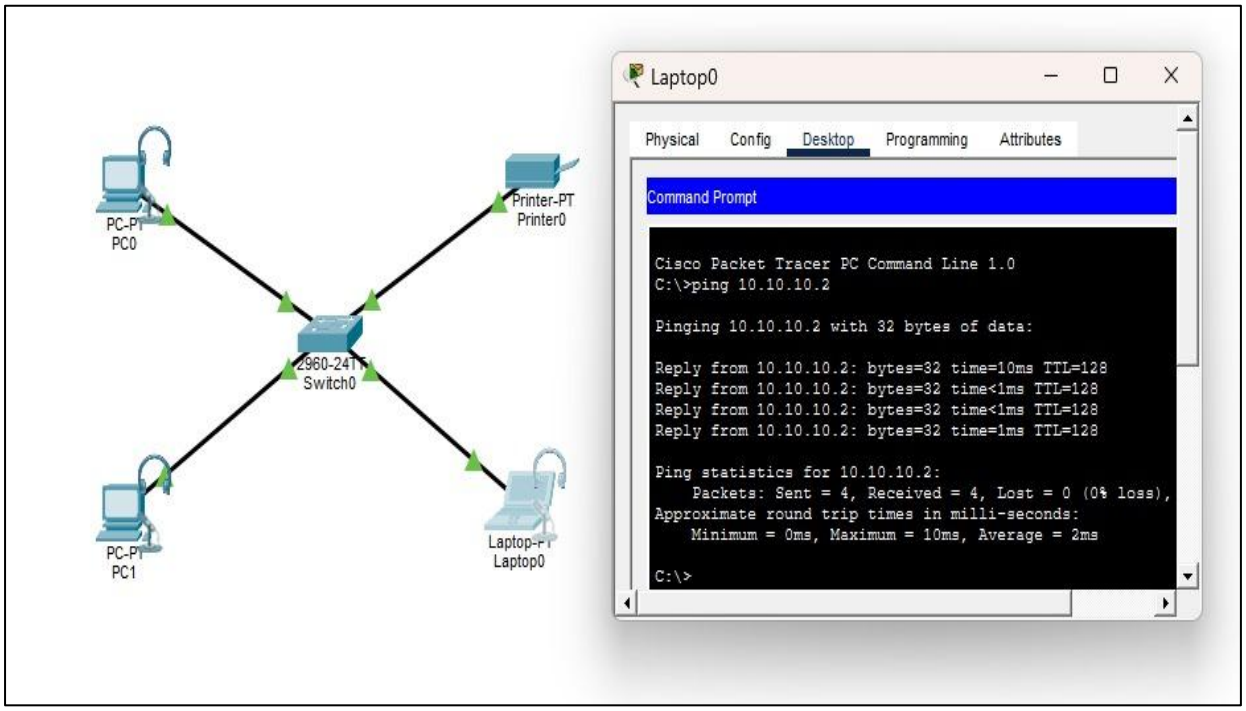


EXP- Message Transfer Between Nodes using Packet Tracer

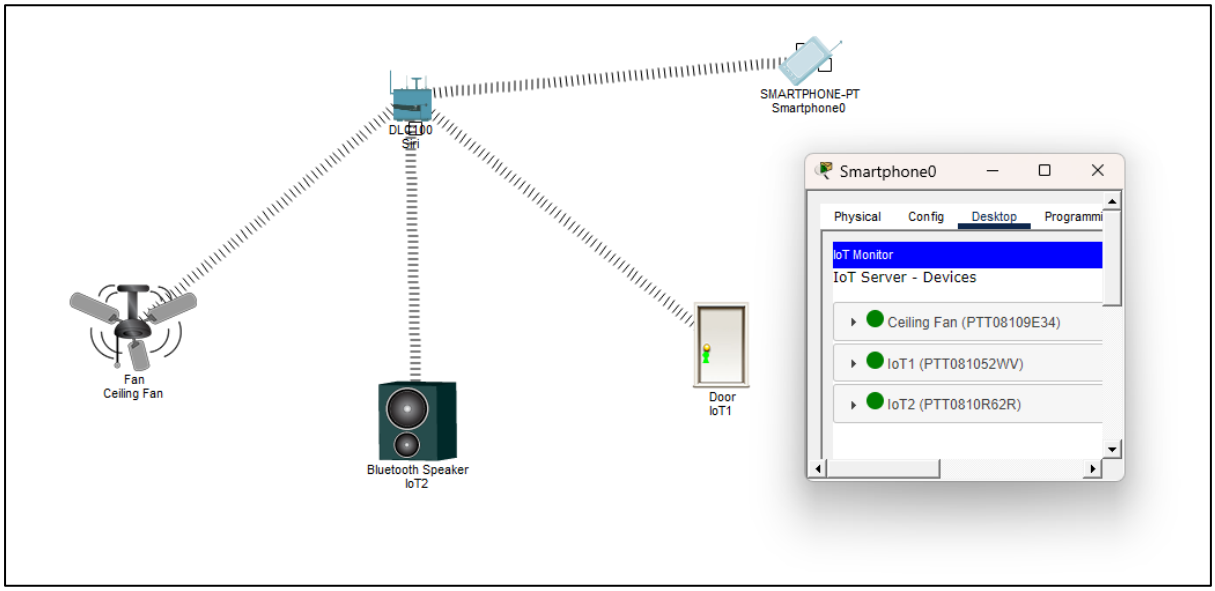




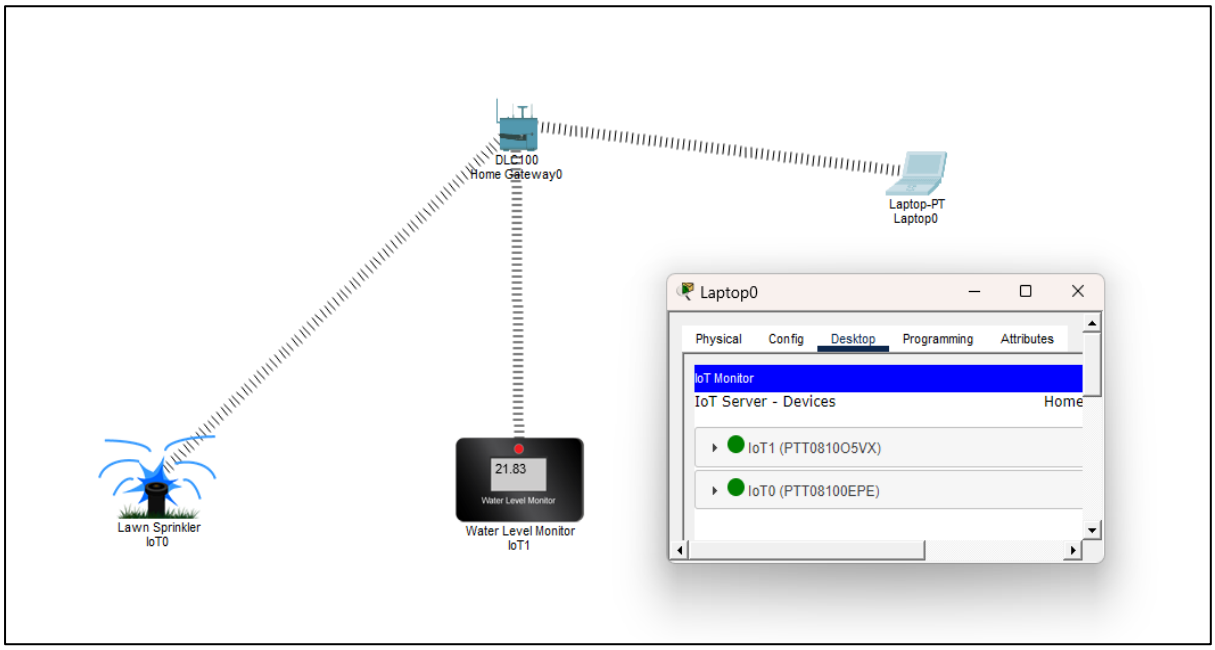
EXP- Simulate a Multimedia Network in Packet Tracer



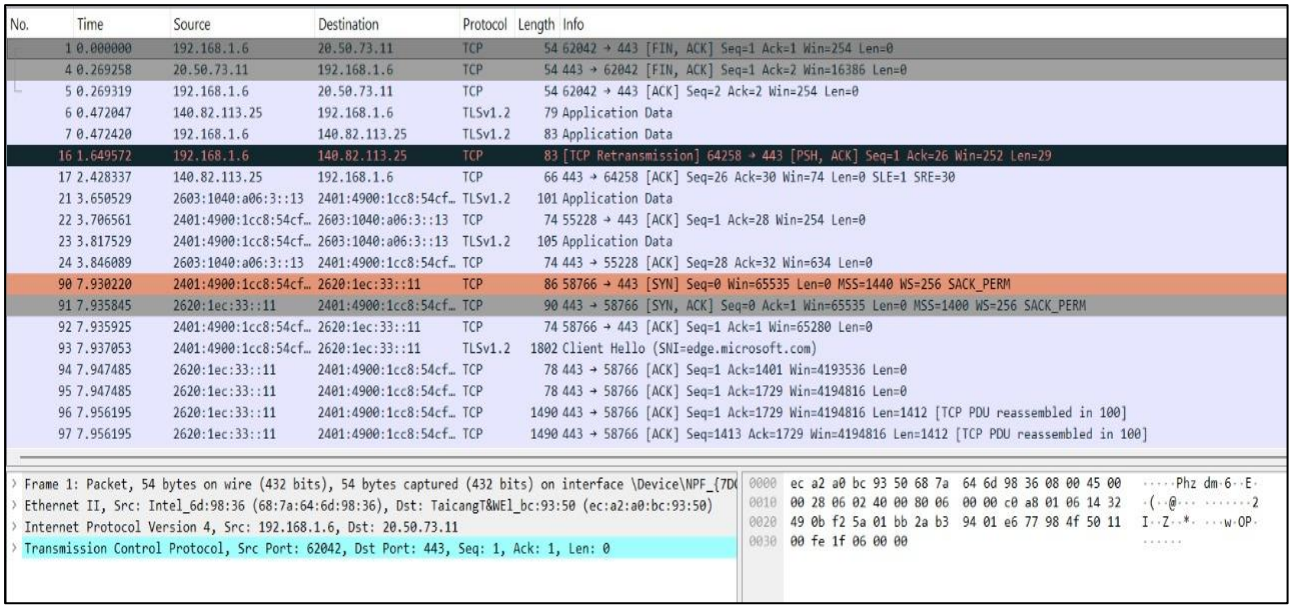
EXP- IoT Based Smart Home Applications



EXP- IoT Based Smart Gardening



EXP- Transport Layer Protocol Header Analysis using Wireshark- TCP & UDP

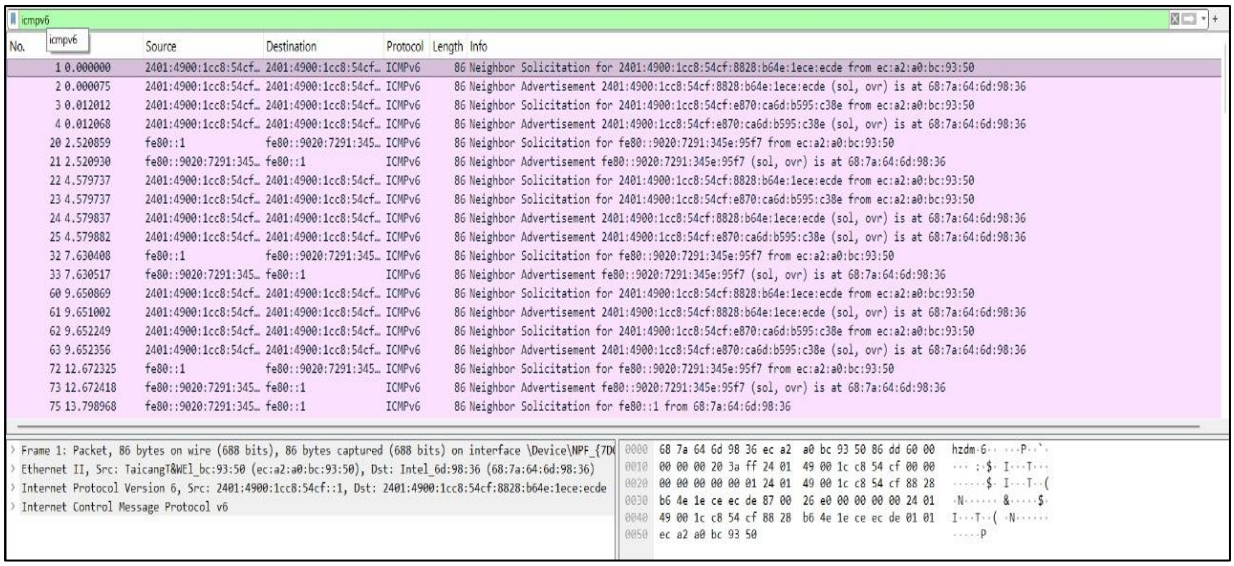


No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.6	20.50.73.11	TCP	54	62042 → 443 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
4	0.269258	20.50.73.11	192.168.1.6	TCP	54	443 → 62042 [FIN, ACK] Seq=1 Ack=2 Win=16386 Len=0
5	0.269319	192.168.1.6	20.50.73.11	TCP	54	62042 → 443 [ACK] Seq=2 Ack=2 Win=254 Len=0
6	0.472047	140.82.113.25	192.168.1.6	TLSv1.2	79	Application Data
7	0.472420	192.168.1.6	140.82.113.25	TLSv1.2	83	Application Data
16	1.649572	192.168.1.6	140.82.113.25	TCP	83	[TCP Retransmission] 64258 → 443 [PSH, ACK] Seq=1 Ack=26 Win=252 Len=29
17	2.428337	140.82.113.25	192.168.1.6	TCP	66	443 → 64258 [ACK] Seq=26 Ack=30 Win=74 Len=0 SLE=1 SRE=30
21	3.650529	2603:1040:a06:3::13	2401:4900:1cc8:54cf…	TLSv1.2	101	Application Data
22	3.706561	2401:4900:1cc8:54cf…	2603:1040:a06:3::13	TCP	74	55228 → 443 [ACK] Seq=1 Ack=28 Win=254 Len=0
23	3.817529	2401:4900:1cc8:54cf…	2603:1040:a06:3::13	TLSv1.2	105	Application Data
24	3.846089	2603:1040:a06:3::13	2401:4900:1cc8:54cf…	TCP	74	443 → 55228 [ACK] Seq=28 Ack=32 Win=634 Len=0
90	7.930220	2401:4900:1cc8:54cf…	2620:1ec:33::11	TCP	86	58766 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
91	7.935845	2620:1ec:33::11	2401:4900:1cc8:54cf…	TCP	90	443 → 58766 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 WS=256 SACK_PERM
92	7.935925	2401:4900:1cc8:54cf…	2620:1ec:33::11	TCP	74	58766 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
93	7.937053	2401:4900:1cc8:54cf…	2620:1ec:33::11	TLSv1.2	1802	Client Hello (SNI=edge.microsoft.com)
94	7.947485	2620:1ec:33::11	2401:4900:1cc8:54cf…	TCP	78	443 → 58766 [ACK] Seq=1 Ack=1401 Win=4193536 Len=0
95	7.947485	2620:1ec:33::11	2401:4900:1cc8:54cf…	TCP	78	443 → 58766 [ACK] Seq=1 Ack=1729 Win=4194816 Len=0
96	7.956195	2620:1ec:33::11	2401:4900:1cc8:54cf…	TCP	1490	443 → 58766 [ACK] Seq=1 Ack=1729 Win=4194816 Len=1412 [TCP PDU reassembled in 100]
97	7.956195	2620:1ec:33::11	2401:4900:1cc8:54cf…	TCP	1490	443 → 58766 [ACK] Seq=1413 Ack=1729 Win=4194816 Len=1412 [TCP PDU reassembled in 100]

```
> Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{7D
> Ethernet II, Src: Intel_6d:98:36 (68:7a:64:6d:98:36), Dst: TaicangT&WEl_bc:93:50 (ec:a2:a0:bc:93:50)
> Internet Protocol Version 4, Src: 192.168.1.6, Dst: 20.50.73.11
> Transmission Control Protocol, Src Port: 62042, Dst Port: 443, Seq: 1, Ack: 1, Len: 0

0000  ec a2 a0 bc 93 50 68 7a  64 6d 98 36 08 00 45 00   ·····Phz dm·6··E·
0010  00 28 06 02 40 00 80 06  00 00 c0 a8 01 06 14 32   ·(··@··· ·******2
0020  49 0b f2 5a 01 bb 2a b3  94 01 e6 77 98 4f 50 11   I··Z··*· ···w·OP·
0030  00 fe 1f 06 00 00                                  ******
```

EXP- Network Layer Protocol Header Analysis using Wireshark- SMTP & ICMP



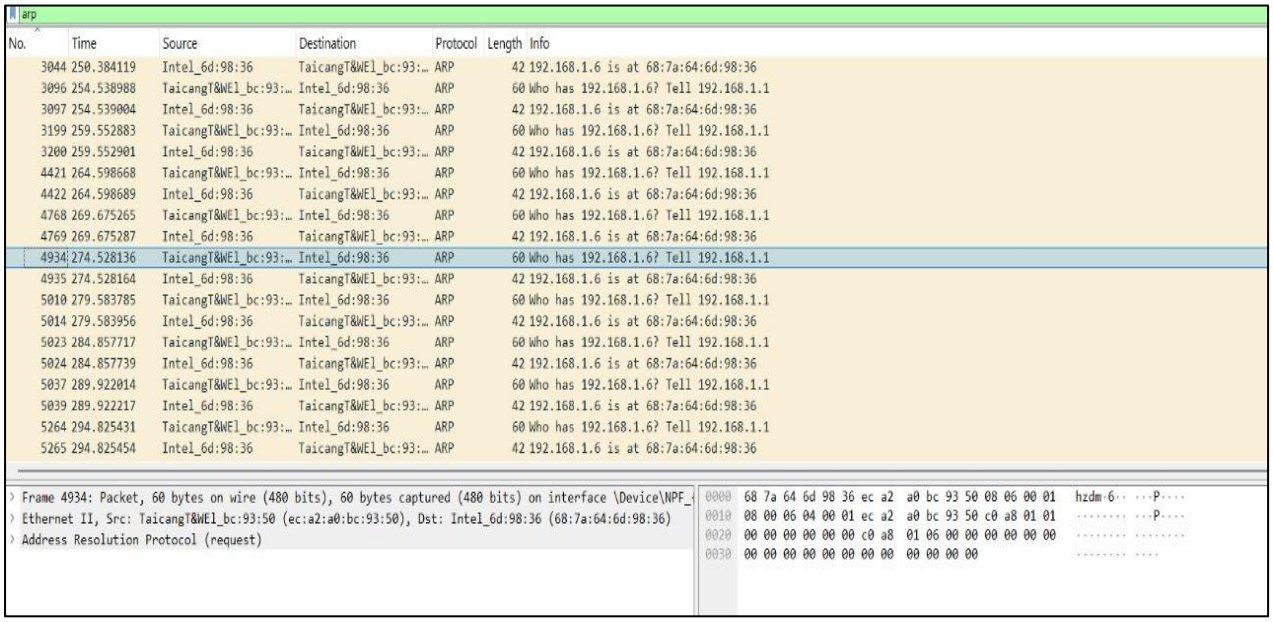
icmpv6

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Solicitation for 2401:4900:1cc8:54cf:8828:b64e:1ece:ecde from ec:a2:a0:bc:93:50
2	0.000075	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Advertisement 2401:4900:1cc8:54cf:8828:b64e:1ece:ecde (sol, ovr) is at 68:7a:64:6d:98:36
3	0.012012	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Solicitation for 2401:4900:1cc8:54cf:e870:ca6d:b595:c38e from ec:a2:a0:bc:93:50
4	0.012068	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Advertisement 2401:4900:1cc8:54cf:e870:ca6d:b595:c38e (sol, ovr) is at 68:7a:64:6d:98:36
20	2.520859	fe80::1	fe80::9020:7291:345…	ICMPv6	86	Neighbor Solicitation for fe80::9020:7291:345e:95f7 from ec:a2:a0:bc:93:50
21	2.520930	fe80::9020:7291:345…	fe80::1	ICMPv6	86	Neighbor Advertisement fe80::9020:7291:345e:95f7 (sol, ovr) is at 68:7a:64:6d:98:36
22	4.579737	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Solicitation for 2401:4900:1cc8:54cf:8828:b64e:1ece:ecde from ec:a2:a0:bc:93:50
23	4.579737	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Solicitation for 2401:4900:1cc8:54cf:e870:ca6d:b595:c38e from ec:a2:a0:bc:93:50
24	4.579837	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Advertisement 2401:4900:1cc8:54cf:8828:b64e:1ece:ecde (sol, ovr) is at 68:7a:64:6d:98:36
25	4.579882	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Advertisement 2401:4900:1cc8:54cf:e870:ca6d:b595:c38e (sol, ovr) is at 68:7a:64:6d:98:36
32	7.630408	fe80::1	fe80::9020:7291:345…	ICMPv6	86	Neighbor Solicitation for fe80::9020:7291:345e:95f7 from ec:a2:a0:bc:93:50
33	7.630517	fe80::9020:7291:345…	fe80::1	ICMPv6	86	Neighbor Advertisement fe80::9020:7291:345e:95f7 (sol, ovr) is at 68:7a:64:6d:98:36
60	9.650869	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Solicitation for 2401:4900:1cc8:54cf:8828:b64e:1ece:ecde from ec:a2:a0:bc:93:50
61	9.651002	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Advertisement 2401:4900:1cc8:54cf:8828:b64e:1ece:ecde (sol, ovr) is at 68:7a:64:6d:98:36
62	9.652249	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Solicitation for 2401:4900:1cc8:54cf:e870:ca6d:b595:c38e from ec:a2:a0:bc:93:50
63	9.652356	2401:4900:1cc8:54cf…	2401:4900:1cc8:54cf…	ICMPv6	86	Neighbor Advertisement 2401:4900:1cc8:54cf:e870:ca6d:b595:c38e (sol, ovr) is at 68:7a:64:6d:98:36
72	12.672325	fe80::1	fe80::9020:7291:345…	ICMPv6	86	Neighbor Solicitation for fe80::9020:7291:345e:95f7 from ec:a2:a0:bc:93:50
73	12.672418	fe80::9020:7291:345…	fe80::1	ICMPv6	86	Neighbor Advertisement fe80::9020:7291:345e:95f7 (sol, ovr) is at 68:7a:64:6d:98:36
75	13.798968	fe80::9020:7291:345…	fe80::1	ICMPv6	86	Neighbor Solicitation for fe80::1 from 68:7a:64:6d:98:36

```
> Frame 1: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{7D
> Ethernet II, Src: TaicangT&WEl_bc:93:50 (ec:a2:a0:bc:93:50), Dst: Intel_6d:98:36 (68:7a:64:6d:98:36)
> Internet Protocol Version 6, Src: 2401:4900:1cc8:54cf::1, Dst: 2401:4900:1cc8:54cf:8828:b64e:1ece:ecde
> Internet Control Message Protocol v6

0000  68 7a 64 6d 98 36 ec a2  a0 bc 93 50 86 dd 60 00   hzdm·6·· ···P··`·
0010  00 00 00 20 3a ff 24 01  49 00 1c c8 54 cf 00 00   ··· :·$· I···T···
0020  00 00 00 00 00 01 24 01  49 00 1c c8 54 cf 88 28   ******$· I···T··(
0030  b6 4e 1e ce ec de 87 00  26 e0 00 00 00 00 24 01   ·N****** &·····$·
0040  49 00 1c c8 54 cf 88 28  b6 4e 1e ce ec de 01 01   I···T··( ·N******
0050  ec a2 a0 bc 93 50                                  ·····P
```

EXP- Network Layer Protocol Header Analysis using Wireshark- ARP & HTTP

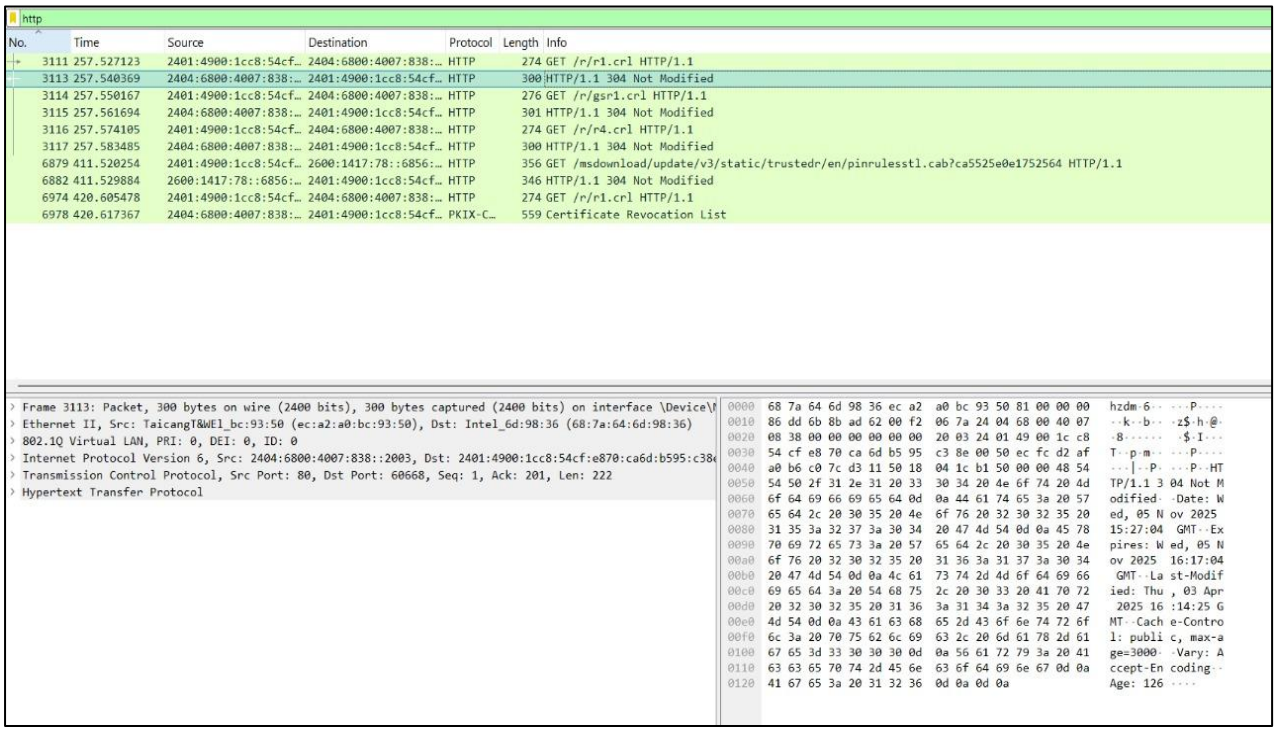


arp

No.	Time	Source	Destination	Protocol	Length	Info
3044	250.384119	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
3096	254.538988	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
3097	254.539004	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
3199	259.552883	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
3200	259.552901	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
4421	264.598668	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
4422	264.598689	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
4768	269.675265	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
4769	269.675287	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
4934	274.528136	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
4935	274.528164	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
5010	279.583785	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
5014	279.583956	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
5023	284.857717	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
5024	284.857739	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
5037	289.922014	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
5039	289.922217	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36
5264	294.825431	TaicangT&WEl_bc:93:…	Intel_6d:98:36	ARP	60	Who has 192.168.1.6? Tell 192.168.1.1
5265	294.825454	Intel_6d:98:36	TaicangT&WEl_bc:93:…	ARP	42	192.168.1.6 is at 68:7a:64:6d:98:36

```
> Frame 4934: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{
> Ethernet II, Src: TaicangT&WEl_bc:93:50 (ec:a2:a0:bc:93:50), Dst: Intel_6d:98:36 (68:7a:64:6d:98:36)
> Address Resolution Protocol (request)

0000  68 7a 64 6d 98 36 ec a2  a0 bc 93 50 08 06 00 01   hzdm·6·· ···P····
0010  08 00 06 04 00 01 ec a2  a0 bc 93 50 c0 a8 01 01   ........ ···P····
0020  00 00 00 00 00 00 c0 a8  01 06 00 00 00 00 00 00   ........ ........
0030  00 00 00 00 00 00 00 00  00 00 00 00               ........ ····
```



http

No.	Time	Source	Destination	Protocol	Length	Info
3111	257.527123	2401:4900:1cc8:54cf…	2404:6800:4007:838:…	HTTP	274	GET /r/r1.crl HTTP/1.1
3113	257.540369	2404:6800:4007:838:…	2401:4900:1cc8:54cf…	HTTP	300	HTTP/1.1 304 Not Modified
3114	257.550167	2401:4900:1cc8:54cf…	2404:6800:4007:838:…	HTTP	276	GET /r/gsr1.crl HTTP/1.1
3115	257.561694	2404:6800:4007:838:…	2401:4900:1cc8:54cf…	HTTP	301	HTTP/1.1 304 Not Modified
3116	257.574105	2401:4900:1cc8:54cf…	2404:6800:4007:838:…	HTTP	274	GET /r/r4.crl HTTP/1.1
3117	257.583485	2404:6800:4007:838:…	2401:4900:1cc8:54cf…	HTTP	300	HTTP/1.1 304 Not Modified
6879	411.520254	2401:4900:1cc8:54cf…	2600:1417:78::6856:…	HTTP	356	GET /msdownload/update/v3/static/trustedr/en/pinrulesstl.cab?ca5525e0e1752564 HTTP/1.1
6882	411.529884	2600:1417:78::6856:…	2401:4900:1cc8:54cf…	HTTP	346	HTTP/1.1 304 Not Modified
6974	420.605478	2401:4900:1cc8:54cf…	2404:6800:4007:838:…	HTTP	274	GET /r/r1.crl HTTP/1.1
6978	420.617367	2404:6800:4007:838:…	2401:4900:1cc8:54cf…	PKIX-C…	559	Certificate Revocation List

```
> Frame 3113: Packet, 300 bytes on wire (2400 bits), 300 bytes captured (2400 bits) on interface \Device\N
> Ethernet II, Src: TaicangT&WEl_bc:93:50 (ec:a2:a0:bc:93:50), Dst: Intel_6d:98:36 (68:7a:64:6d:98:36)
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
> Internet Protocol Version 6, Src: 2404:6800:4007:838::2003, Dst: 2401:4900:1cc8:54cf:e870:ca6d:b595:c38
> Transmission Control Protocol, Src Port: 80, Dst Port: 60668, Seq: 1, Ack: 201, Len: 222
> Hypertext Transfer Protocol

0000  68 7a 64 6d 98 36 ec a2  a0 bc 93 50 81 00 00 00   hzdm·6·· ···P····
0010  86 dd 6b 8b ad 62 00 f2  06 7a 24 04 68 00 40 07   ··k··b·· ·z$·h·@·
0020  08 38 00 00 00 00 00 00  20 03 24 01 49 00 1c c8   ·8····· ·$·I····
0030  54 cf e8 70 ca 6d b5 95  c3 8e 00 50 ec fc d2 af   T··p·m·· ···P····
0040  a0 b6 c0 7c d3 11 50 18  04 1c b1 50 00 00 48 54   ···|··P· ···P··HT
0050  54 50 2f 31 2e 31 20 33  30 34 20 4e 6f 74 20 4d   TP/1.1 3 04 Not M
0060  6f 64 69 66 69 65 64 0d  0a 44 61 74 65 3a 20 57   odified· ·Date: W
0070  65 64 2c 20 30 35 20 4e  6f 76 20 32 30 32 35 20   ed, 05 N ov 2025
0080  31 35 3a 32 37 3a 30 34  20 47 4d 54 0d 0a 45 78   15:27:04  GMT··Ex
0090  70 69 72 65 73 3a 20 57  65 64 2c 20 30 35 20 4e   pires: W ed, 05 N
00a0  6f 76 20 32 30 32 35 20  31 36 3a 31 37 3a 30 34   ov 2025  16:17:04
00b0  20 47 4d 54 0d 0a 4c 61  73 74 2d 4d 6f 64 69 66    GMT··La st-Modif
00c0  69 65 64 3a 20 54 68 75  2c 20 30 33 20 41 70 72   ied: Thu , 03 Apr
00d0  20 32 30 32 35 20 31 36  3a 31 34 3a 32 35 20 47    2025 16 :14:25 G
00e0  4d 54 0d 0a 43 61 63 68  65 2d 43 6f 6e 74 72 6f   MT··Cach e-Contro
00f0  6c 3a 20 70 75 62 6c 69  63 2c 20 6d 61 78 2d 61   l: publi c, max-a
0100  67 65 3d 33 30 30 0d 0a  56 61 72 79 3a 20 41 63   ge=3000· ·Vary: A
0110  63 65 70 74 2d 45 6e 63  6f 64 69 6e 67 0d 0a      ccept-En coding··
0120  41 67 65 3a 20 31 32 36  0d 0a 0d 0a               Age: 126 ····
```